Low-Cost Optical Sensor System for Microplastic Detection in Water

A PROJECT REPORT

Submitted by

SANJEEVRAJ BOMMANNAN M

24ECR174

NANDITA M K

24ECR137

SANJAY B

24ECR172

in partial fulfilment of the requirements

for the award of the degree

of

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



KONGU ENGINEERING COLLEGE

(Autonomous)

PERUNDURAI ERODE – 638 052

APRIL 2026

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING KONGU ENGINEERING COLLEGE

(Autonomous)
PERUNDURAI ERODE – 638052

APRIL 2026

BONAFIED CERTIFICATE

This is to certify that the Mini Project report entitled **Low-Cost Optical Sensor System for Microplastic Detection in Water** is the Bonafide record of project work done by **SANJEEVRAJ BOMMANNAN M 24ECR174 , NANDITA M K 24ECR137 , SANJAY B 24ECR172** in partial fulfilment of the requirements for the award of the Degree of Bachelor of Engineering in **ELECTRONICS AND COMMUNICATION** of Anna university Chennai during the year 2025 - 2026.

SUPERVISOR
Mr. S. SUTHAGAR M.Tech.,
Assistant Professor,
Department of ECE,
Kongu Engineering College,
Perundurai – 638060.

HEAD OF THE DEPARTMENT
Dr. N. KASTHURI M.E., PhD.,
Professor and Head,
Department of ECE,
Kongu Engineering College,
Perundurai – 638060.

Date:

Submitted for the end semester viva voce examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING KONGU ENGINEERING COLLEGE

(Autonomous)
PERUNDURAI ERODE – 638052

APRIL 2026

DECLARATION

We affirm that the Mini Project Report titled **Low-Cost Optical Sensor System for Microplastic Detection in Water** being submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

(Signature of the candidate)

Date:

SANJEEVRAJ BOMMANNAN M
(24ECR174)

NANDITA M K
(24ECR137)

SANJAY B
(24ECR172)

I certify that the declaration made by the above candidates is true to the best of my knowledge.

Name and Signature of the Supervisor with seal

Date:

ABSTRACT

Microplastic contamination in water has emerged as a serious environmental and public health concern due to the widespread use of plastic materials and their slow degradation into particles smaller than 5 mm. These particles enter water bodies through domestic waste, industrial discharge, and environmental degradation of plastic products, posing significant risks to aquatic ecosystems and human health. Conventional microplastic detection techniques such as Raman spectroscopy, FTIR analysis, and fluorescence imaging provide high accuracy but require expensive laboratory equipment, complex sample preparation, and skilled operation. These limitations make them unsuitable for real-time field monitoring and low-cost deployment.

This project proposes a low-cost optical sensor system for microplastic detection in water using multi-wavelength light sources and photodiode sensors. The system utilizes UV, visible, and near-infrared (NIR) LEDs to illuminate the water sample, while photodiodes positioned at different angles measure transmission and scattering characteristics. Variations in light intensity caused by the presence of microplastic particles are analyzed to detect their presence and estimate concentration. The collected signals are processed using an embedded microcontroller (ESP32), enabling real-time analysis and portable operation.

The main advantages of this system include low implementation cost, portability, real-time monitoring capability, and ease of operation. However, the system may have limitations in identifying exact polymer types compared to advanced spectroscopic techniques and may require calibration for different water conditions. Overall, this work contributes toward the development of an affordable and practical microplastic monitoring solution that can be used for environmental monitoring, water quality assessment, and preliminary screening applications.

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SUSTAINABLE DEVELOPMENT GOALS

This project on microplastic detection in water using a low-cost optical sensor system supports several United Nations Sustainable Development Goals (SDGs), particularly SDG 6 (Clean Water and Sanitation), SDG 12 (Responsible Consumption and Production), and SDG 14 (Life Below Water). By enabling early detection of microplastic contamination, the system contributes to improving water quality and promotes safer water resources for both human use and aquatic ecosystems. The project also creates awareness about the harmful effects of plastic pollution and encourages responsible waste management practices. Furthermore, by helping to monitor pollutants that threaten marine and freshwater life, the system supports environmental protection and biodiversity conservation. Due to its low cost, portability, and real-time monitoring capability, the proposed solution can be widely used for environmental screening and research applications, thereby contributing toward sustainable environmental monitoring and pollution control. Overall, the project aligns with global sustainability efforts by providing a practical technological solution to address the growing problem of microplastic pollution.



OBJECTIVE	**To develop a low-cost optical sensor system for real-time detection and estimation of microplastic contamination in water**

ACKNOWLEDGEMENT

First and foremost, we acknowledge the abundant grace and presence of the almighty throughout different phases of the project and its successful completion.

We wish to express our hearty gratitude to our honorable Correspondent **Thiru. E. R. K. KRISHNAN M.Com.,** and other trust members for having provided us with all the necessary infrastructure to undertake this project.

We extend our hearty gratitude to our honorable Principal **Dr. R. PARAMESHWARAN M.E., Ph.D.,** for his consistent encouragement throughout our college days.

We would like to express our profound interest and sincere gratitude to our respected Head of the department **Dr. N. KASTHURI ME., Ph.D.,** for her valuable guidance.

A special debt is owed to the project coordinators **Dr. MALATHI D M.E., Ph.D.,** Associate Professor & **Mr. S. SUTHAGAR., M.Tech.,** Assistant Professor Department of Electronics and Communication Engineering for their encouragement and valuable advice that made us carry out the project work successfully.

We extend our sincere gratitude to our beloved guide **Mr. S. SUTHAGAR., M.Tech.,** Assistant Professor, Department of Electronics and Communication Engineering for her ideas and suggestions, which have been very helpful to complete the project.

We are grateful to all the faculty and staff members of the Electronics and Communication Engineering Department and persons who directly and indirectly supported this project

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CHAPTER – 1

INTRODUCTION

1.1 INTRODUCTION

Microplastics are plastic particles smaller than 5 mm that are increasingly found in oceans, rivers, groundwater, and drinking water sources. These particles are mainly produced from the degradation of plastic waste, industrial discharge, synthetic textiles, and domestic products. Due to their very small size, microplastics cannot be easily removed by conventional filtration systems and may enter the food chain, posing serious risks to aquatic organisms and human health.

Traditional microplastic detection methods such as Raman spectroscopy, FTIR spectroscopy, and electron microscopy provide high accuracy but require expensive laboratory equipment, skilled operators, and long processing time. These limitations make them unsuitable for real-time and field applications.

Recent research focuses on optical sensing techniques because they provide a fast, non-destructive, and cost-effective solution. Optical detection works based on light interaction with particles such as absorption, scattering, and transmission variation. When light passes through contaminated water, the intensity changes depending on the presence of particles.

This project proposes a Low-Cost Optical Sensor System for Microplastic Detection in Water using multi-wavelength LEDs (UV, Visible, and IR) and BPW34 photodiode sensors. The ESP32 microcontroller is used to process the optical signals and determine whether microplastics are present.

1.2 PROBLEM STATEMENT

Microplastic pollution is increasing rapidly due to excessive plastic usage and improper waste management. Monitoring microplastics is important, but existing detection methods have several drawbacks.

Major problems include:

• Laboratory-dependent detection methods such as Raman and FTIR spectroscopy are very expensive.

• Most existing systems cannot be used for field monitoring.

• Low-cost sensors like turbidity sensors cannot differentiate microplastics from dust or organic particles.

• Existing portable systems cannot estimate concentration accurately.

• No compact embedded system provides real-time optical analysis using low-cost components.

Because of these limitations, there is a strong need for a low-cost embedded microplastic detection system that can provide real-time monitoring using simple optical components.

This project aims to solve these problems by developing a compact optical sensing system using LEDs, photodiodes, and ESP32 processing.

1.3 OBJECTIVE

The main objective of this project is to develop a low-cost optical sensing system to detect microplastics in water using embedded hardware.

Specific objectives include:

• To design a low-cost optical detection system using LEDs and photodiodes.

• To detect microplastic presence using transmission and scattering principles.

• To compare pure water and contaminated water optical characteristics.

• To process sensor data using ESP32 microcontroller.

• To estimate concentration variation based on optical intensity difference.

• To develop a compact prototype for real-time monitoring.

• To create a portable environmental testing device.

CHAPTER – 2

LITERATURE REVIEW

2.1 INTRODUCTION

Microplastic detection has gained significant attention due to increasing environmental pollution and health risks. Various techniques such as spectroscopy, optical sensing, fluorescence imaging, and machine learning-based detection have been proposed by researchers. Each method provides advantages in detection accuracy, automation, or real-time monitoring but also suffers from limitations such as high cost, complex setup, or lack of portability.

Traditional laboratory techniques such as Raman spectroscopy provide accurate polymer identification but require expensive instruments and controlled environments. Recent studies focus on low-cost optical detection using LEDs and photodiodes for embedded implementation.

This chapter reviews important research works related to microplastic detection and discusses their advantages and limitations.

2.2 OPTICAL SCATTERING BASED DETECTION

Zhang et al. developed an optical scattering-based detection system that measures the scattering intensity variations caused by microplastic particles [1]. Their system demonstrated high sensitivity and accurate detection capability.

However, the system required expensive optical components and laboratory calibration, limiting its portability. This study shows that optical scattering is effective but requires cost optimization.

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2.3 EMBEDDED OPTICAL MONITORING SYSTEMS

Kumar and Lee proposed an embedded real-time microplastic monitoring system using LED and photodiode sensors [2]. Their system provided portable detection and low power consumption.

However, the system could only detect the presence of particles and could not accurately classify polymer types. This shows that embedded optical systems are suitable for low-cost detection but require further improvements.

2.4 IoT BASED MICROPLASTIC MONITORING

Sharma et al. developed an IoT-based optical water quality monitoring system including microplastic detection [3]. The system enabled real-time data transmission and monitoring.

Although the system provided continuous monitoring, it lacked high selectivity and required internet connectivity. This indicates the need for simpler standalone embedded systems.

2.5 MACHINE LEARNING BASED DETECTION

Cerioli et al. developed a TinyML-based optical classification system to detect microplastics using embedded machine learning [4]. Their system achieved low power processing and improved detection accuracy.

However, the system required trained datasets and complex model development. This shows the potential of ML but also highlights implementation complexity.

Similarly, Patel et al. proposed a deep learning-based microscopic classification system [8]. Their system achieved good classification accuracy but required high computational resources.

2.6 MICROWAVE AND SPECTROSCOPY METHODS

Zhao et al. developed a microwave resonator-based microplastic detection system capable of concentration estimation [5]. However, the system could not identify polymer types.

Liu et al. used Raman spectroscopy for polymer identification [6]. Their system achieved high accuracy but required expensive instrumentation.

Lu et al. proposed a portable SERS-based nanoplastic detection system capable of detecting nano-scale plastics [7]. However, the system required costly substrates.

These studies show that spectroscopy methods provide high accuracy but are expensive.

2.7 FLUORESCENCE AND IMAGING METHODS

Vitali et al. developed a fluorescence imaging system combined with AI for microplastic quantification [9]. The system provided good concentration estimation but required chemical staining.

Passananti et al. proposed a flow cytometry-based rapid microplastic analysis method [10]. The system provided fast detection but had limited field deployment capability.

Kim et al. proposed UV-induced fluorescence detection of microplastics [11]. The system showed polymer sensitivity but required controlled testing conditions.

2.8 ADVANCED OPTICAL SENSOR METHODS

Xiong et al. developed an optical fiber-based nanosensor for detecting plastic particles [12]. The system detected nanoplastics but was limited to specific polymer types.

Garcia et al. developed a portable microplastic analyzer using optical imaging [13]. The system provided moderate accuracy but required image processing.

Brown et al. proposed multi-spectral optical fingerprinting for microplastic detection [14]. Their system provided spectral analysis but required complex hardware.

Ahmed et al. developed a laser-induced spectral analysis method for microplastic detection [15]. The system provided high accuracy but required expensive laser setups.

2.9 RESEARCH GAP AND SUMMARY

The literature review shows that many techniques have been developed for detecting microplastics using optical sensing, spectroscopy, imaging, and machine learning methods. Each method provides certain advantages but also has limitations related to cost, portability, complexity, and real-time implementation.

Spectroscopy methods such as Raman and SERS provide high accuracy in polymer identification but require expensive laboratory instruments and skilled operation [6][7].

Optical LED-photodiode systems provide low-cost detection and portability but have limited capability in polymer classification [1][2]. IoT-based monitoring systems enable real-time data monitoring but increase system complexity and depend on continuous internet connectivity [3].

Machine learning based detection methods improve classification accuracy but require large training datasets and higher computational resources [4][8]. Fluorescence and imaging methods provide good concentration estimation but require chemical preparation and complex image processing [9][13]. Advanced optical techniques such as laser spectral analysis provide high precision but increase overall system cost [14][15].

From this review, it is clear that no existing system provides a combination of:

• Low-cost hardware implementation

• Portable embedded system design

• Multi-wavelength optical detection

• Real-time analysis capability

• Simple and compact implementation

Therefore, there is a clear research gap for developing a low-cost, portable, and real-time microplastic detection system using simple optical components and embedded processing.

The proposed project addresses this gap by developing a microplastic detection system using multi-wavelength LEDs (UV, Visible, IR), BPW34 photodiode sensors, and ESP32 microcontroller processing. The system focuses on detecting microplastics based on transmission and scattering variations while maintaining low cost and portability.

This chapter reviewed the existing techniques and identified the need for an economical and embedded optical detection system. The next chapter explains the existing methods and the proposed methodology of the system.

CHAPTER – 3

EXISTING METHODOLOGY

3.1 INTRODUCTION

Microplastic detection is commonly performed using spectroscopy, optical imaging, fluorescence analysis, and optical scattering techniques. These methods mainly focus on identifying plastic particles based on their chemical, optical, and physical properties. Many research works have demonstrated high detection accuracy using these techniques. However, most of these systems require expensive laboratory equipment and complex processing. This section discusses some important existing methods along with their advantages and limitations based on previous research papers.

3.2 RAMAN SPECTROSCOPY METHOD

Liu et al. proposed a Raman spectroscopy-based method for identifying polymer types of microplastics by analyzing molecular vibration signatures [6]. This method uses laser excitation to obtain spectral fingerprints of plastic materials, enabling accurate identification of different polymer types.

Advantages:

- High accuracy in polymer identification
- Non-destructive testing method
- Capable of identifying very small particles
- Reliable material classification

Limitations:

- Very expensive instrumentation
- Requires laboratory setup
- Needs skilled operators
- Not suitable for portable applications

3.3 OPTICAL SCATTERING METHOD

Zhang et al. developed an optical scattering-based detection system where light scattering variation is used to detect microplastic particles [1]. The system measures scattering intensity changes when particles interact with light.

Advantages:

- High sensitivity detection
- Non-contact measurement
- Fast detection process
- Good for particle presence detection

Limitations:

- Requires precise optical alignment
- Expensive optical components
- Requires calibration
- Limited polymer identification capability

3.4 MACHINE LEARNING BASED OPTICAL DETECTION

Cerioli et al. proposed a TinyML-based optical signal classification system to detect microplastics using embedded machine learning [4]. The system processes optical signals and classifies microplastic presence using trained models.

Advantages:

- Improved detection accuracy
- Low power embedded processing
- Real-time classification possible
- Scalable approach

Limitations:

- Requires training dataset
- Model development complexity
- Requires optimization
- Limited without proper training data

3.5 FLUORESCENCE IMAGING METHOD

Vitali et al. proposed a fluorescence imaging system combined with artificial intelligence to quantify microplastic concentration [9]. The method uses fluorescent dyes to highlight microplastic particles under specific wavelength illumination.

Advantages:

• Good concentration estimation

• Enhanced visibility of particles

• Suitable for research analysis

• Can detect small particles

Limitations:

• Requires chemical staining

• Requires imaging equipment

• Complex processing

3.6 COMPARATIVE ANALYSIS OF EXISTING SYSTEMS

Method	Methodology	Technology Used	Advantages	Limitations
Raman Spectroscopy Method	Uses laser light interaction to identify molecular structure of plastics	Laser source, spectrometer, optical filters	Very high accuracy, polymer identification possible	Very expensive, laboratory dependent, complex operation
Optical Scattering Method	Measures light scattering variation when particles interact with light	LED/Laser source, photodetector, optical setup	Fast detection, good sensitivity, non-destructive	Requires calibration, limited polymer classification
Machine Learning Based Optical Detection	Uses optical signal data and ML algorithms for classification	Optical sensors, microcontroller, ML models	Improved detection accuracy, automation possible	Requires training data, complex model development
Fluorescence Imaging Method	Uses fluorescent dyes and UV light to visualize microplastics	UV light source, microscope, imaging system	Good particle visibility, concentration estimation	Requires chemical staining, expensive imaging setup

Table 3.6.1

From this comparison, Raman spectroscopy provides high accuracy but is expensive and laboratory dependent. Optical scattering provides a simple detection method but lacks material classification capability. Machine learning-based detection improves accuracy but increases system complexity. Fluorescence imaging provides good visualization but requires chemical preparation. Therefore, there is a need for a low-cost, portable optical detection system using simple LEDs and photodiode sensors with embedded processing, which is addressed by the proposed system..

3.7 SUMMARY

This chapter discussed the existing methodologies used for microplastic detection such as Raman spectroscopy, optical scattering methods, machine learning-based optical detection, and fluorescence imaging techniques. Raman spectroscopy provides highly accurate polymer identification but requires expensive laboratory equipment. Optical scattering methods provide fast detection but have limited capability in material classification. Machine learning-based optical detection improves detection accuracy but requires training datasets and higher computational resources. Fluorescence imaging methods provide good particle visualization but require chemical staining and complex imaging setups.

From this study, it is observed that most existing systems either provide high accuracy with high cost or low cost with limited detection capability. None of the systems provide a balance between low cost, portability, and real-time detection using simple embedded hardware.

Therefore, there is a need for a compact, low-cost optical detection system that can detect microplastics using simple optical components and embedded processing. The next chapter explains the proposed methodology used in this project to overcome these limitations by using multi-wavelength LEDs, photodiode sensors, and ESP32 microcontroller processing.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 INTRODUCTION

The proposed system is designed to detect microplastics in water using optical sensing principles. The system uses a low-cost embedded approach instead of expensive laboratory techniques. The detection is based on the interaction between light and microplastic particles, where the presence of particles affects light transmission, absorption, and scattering characteristics. By measuring these optical variations, the system determines whether microplastics are present in the water sample.

The proposed system uses multi-wavelength LEDs such as UV, Visible (white), and IR LEDs as light sources and a BPW34 photodiode as the optical detector. The ESP32 microcontroller is used to process the sensor signals and determine the presence of microplastics. The system is designed to be compact, portable, and suitable for real-time monitoring.

4.2SYSTEM ARCHITECTURE

The system consists of a multi-wavelength LED array, a water sample chamber, a photodiode sensor, a signal conditioning circuit, and an ESP32 microcontroller. The LEDs are used to illuminate the water sample from different wavelengths. The photodiode is placed to receive transmitted and scattered light from the sample. The photodiode converts light intensity into electrical signals.

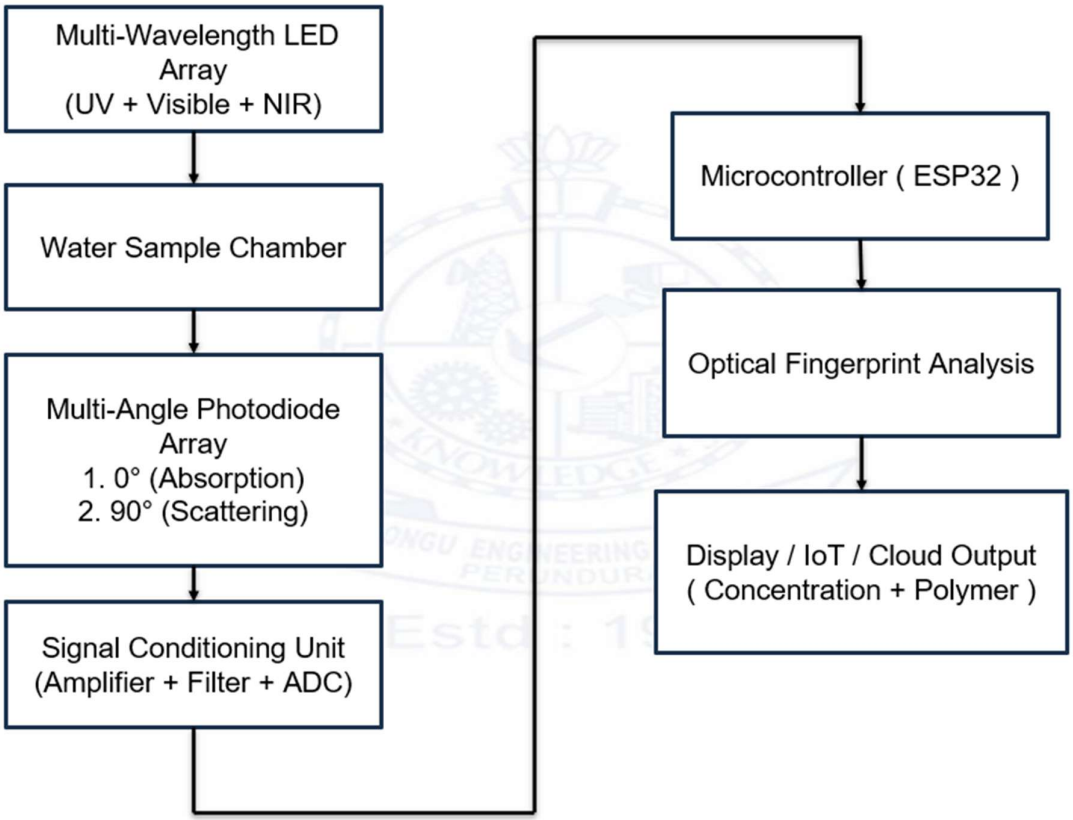
Since the photodiode output voltage is very small, an LM358 operational amplifier is used to amplify the signal. The amplified signal is then given to the ESP32 analog input pins. The ESP32 processes the sensor readings and compares them with reference values obtained from pure water. Based on the difference in readings, the system determines whether microplastics are present.

4.3 WORKING PRINCIPLE

The working principle of the proposed system is based on optical transmission and scattering measurement. When light passes through pure water, most of the light is transmitted because there are no particles to obstruct the path. When microplastics are present, they cause scattering and absorption of light, which reduces transmission intensity and increases scattered light intensity.

The system first measures the empty container reading to establish the baseline value. Then pure water is tested to obtain the reference reading. Finally, the contaminated water sample is tested. The ESP32 compares the pure water value and the sample value. If the difference exceeds a predefined threshold, the system indicates the presence of microplastics. This comparison method improves detection reliability.

4.4 BLOCK DIAGRAM



4.5 COMPONENTS USED

4.5.1 ESP32 Microcontroller

The ESP32 is the main processing unit of the system. It is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities. In this project, the ESP32 reads the photodiode sensor values, processes the data, and determines the presence of microplastics based on intensity variations.

Features and Specifications

- Dual-core 32-bit Xtensa LX6 microprocessor
- Operating frequency: 160 MHz to 240 MHz
- Built-in Wi-Fi 802.11 b/g/n and Bluetooth v4.2 (BR/EDR + BLE)
- Flash memory: 4 MB
- SRAM: 520 KB
- Operating voltage: 3.0 V to 3.6 V
- 30 to 36 GPIO pins
- ADC (12-bit) and DAC support
- UART, SPI, I2C, PWM interfaces available
- Power consumption (Active mode): < 240 mA



Figure 4.5.1**:** ESP32 Microcontroller

4.5.2 BPW34 PHOTODIODE

The BPW34 photodiode is used to detect the light intensity after interaction with the water sample. It converts the incident light into a proportional electrical current which is used to analyze transmission and scattering variations. Due to its wide spectral response range (400–1100 nm) and fast response time, it is suitable for optical sensing applications.

Features:

The BPW34 photodiode provides high sensitivity and low noise performance, making it suitable for accurate optical measurements. It also offers fast response and reliable detection, which improves measurement stability in the microplastic detection system.

Specifications:

Parameter	Specification
Spectral Range	400–1100 nm
Peak Sensitivity	900 nm
Operating Voltage	0–60V
Response Time	Fast
Package Type	Through hole

Table 4.5.2

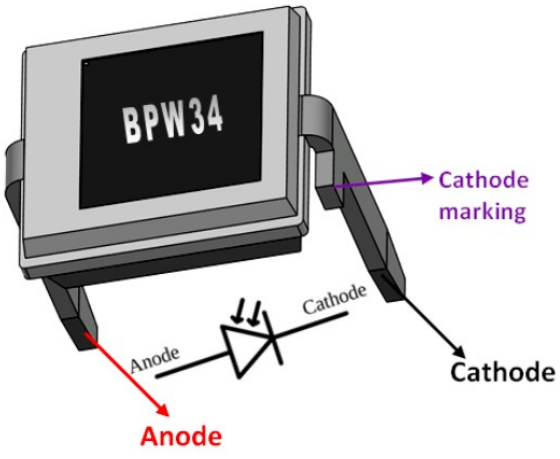


Figure 4.5.2: BPW34 PHOTO DIODE

4.5.3 UV LED

The UV LED is used to analyze fluorescence characteristics of microplastic particles. It emits ultraviolet light which interacts with plastic particles and helps identify optical variations.

Features

The UV LED provides stable ultraviolet output and low power consumption. It is suitable for fluorescence-based detection and improves optical analysis capability.

Specifications

Parameter	**Specification**
Wavelength	365 nm
Forward Voltage	3–3.4V
Current Rating	20mA
LED Type	5mm

Table 4.5.3



Figure 4.5.3: UV LED

4.5.4 IR LED

The IR LED is used to measure transmission characteristics of the water sample. It helps in identifying absorption variations caused by microplastic particles.

Functions

- Good penetration capability
- Stable infrared emission
- Low cost component
- Suitable for transmission measurement

Specifications

Parameter	Specification
Wavelength	**850 nm**
Forward Voltage	**1.2–1.5V**
Current	**20–50mA**
LED Type	**5mm**

Table 4.5.4



Figure 4.5.4: IR LED

4.5.5 WHITE LED

The white LED is used to measure scattering effects in the water sample. When microplastics are present, scattering increases which affects detected intensity.

Features

• High brightness output

• Stable light emission

• Low power requirement

Specification:

Parameter	Specification
Forward Voltage	**3V**
Current	**20mA**
Light Type	**Visible white**
LED Size	**5mm**

Table 4.5.4



Figure 4.5.5: IR LED

4.5.6 LM358 OPERATIONAL AMPLIFIER

The LM358 operational amplifier is used to amplify the small electrical signal produced by the photodiode. Since the photodiode output is very weak, amplification is necessary before giving it to the ESP32 ADC.

Features:

- Low power operation
- Dual amplifier package
- Stable signal amplification
- Wide voltage operating range
- Suitable for sensor signal conditioning

Specifications:

Parameter	Specification
Supply Voltage	3–32V
Number of Op-Amps	2
Gain Bandwidth	1 MHz
Input Offset Voltage	Low

Table 4.5.6

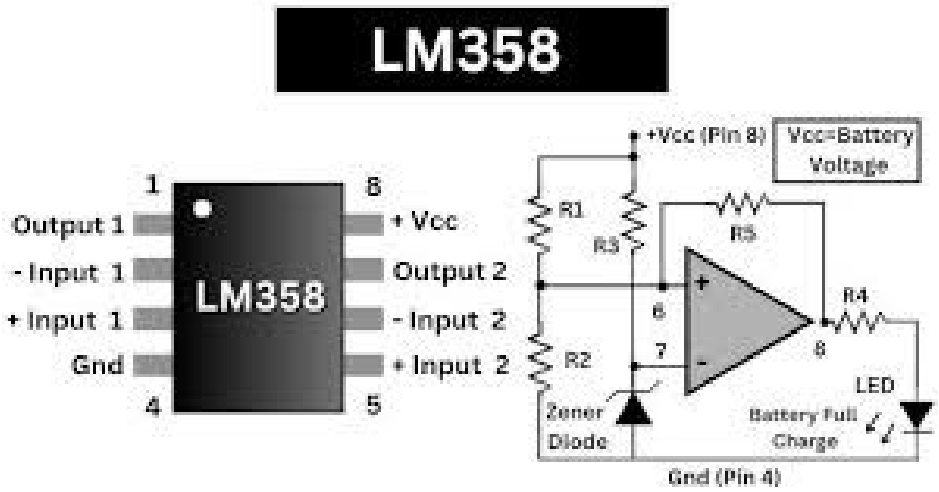


Figure 4.5.6: LM358

4.5.7 WATER SAMPLE CHAMBER

The water sample chamber is used to hold the test sample during optical testing. It is made of transparent acrylic or glass to allow proper light transmission.

Features:

• High optical transparency

• Low cost container

• Easy sample handling

• Reusable structure

• Suitable for optical experiments

Specification:

Parameter	**Specification**
Material	**Acrylic / Glass**
Width	**~6–10 cm**
Transparency	**High**
Type	**Static container**

Table 4.5.7



Figure 4.5.7: LM358

4.6 COST ANALYSIS

S.No	Component	Specification	Purpose	Approx Cost (₹)
1	UV LED	365 nm, 5mm	Fluorescence testing	12
2	White LED	5mm, 20mA	Scattering measurement	10
3	IR LED	850 nm	Transmission analysis	10
4	BPW34 Photodiode	400–1100 nm	Optical detection	100
5	LM358 Op-Amp	Dual Op-Amp	Signal amplification	20
6	ESP32	Dual core MCU	Processing unit	350
7	Sample Chamber	Acrylic container	Water testing	100
8	Breadboard & Wires	Standard	Circuit setup	40
9	Passive Components	Resistors, capacitors	Circuit support	30

Table 4.6.1

Total Estimated Cost ≈ ₹800
The cost may vary depending on component availability and quality.

4.7 ADVANTAGES OF THE PROPOSED SYSTEM

• Low-cost system compared to Raman and FTIR spectroscopy methods

• Portable design suitable for field testing and environmental monitoring

• Real-time detection without laboratory dependency

• Simple hardware implementation using LEDs and photodiode

• Easy calibration and maintenance

• No complex sample preparation required

4.8 SUMMARY

This chapter explained the proposed methodology, system architecture, working principle, and hardware components used in the microplastic detection system. The specifications and features of the components were discussed along with the cost analysis of the prototype. The proposed system provides a simple, economical, and portable solution for detecting microplastics using optical sensing techniques.

CHAPTER – 5
RESULT AND DISCUSSION

5.1 RESULT

The proposed microplastic detection system was tested using different samples such as empty container, pure water, and contaminated water sample. The readings from the photodiode sensor were recorded for UV, White, and IR light sources. The difference between the reference readings and sample readings was used to determine the presence of microplastics.

The experimental results showed that pure water produced higher transmission values and lower scattering values compared to contaminated samples. When microplastic particles were present, the transmission value decreased and scattering increased due to light interaction with particles. The ESP32 successfully detected these variations and displayed the detection result through the serial monitor.

Observation Table

Sample	**UV Transmission**	**White Scattering**	**IR Transmission**	**Result**
Empty Container	120	5600	4800	Reference
Pure Water	110	6000	4700	Clean Water
Sample Water	70	7500	4000	Microplastic Detected

5.2 DISCUSSION

From the experimental results, it is observed that the optical properties of water change when microplastic particles are present. The reduction in transmission values and increase in scattering values confirms the presence of suspended particles. The UV LED provided noticeable variation due to fluorescence interaction, while the white LED showed scattering changes.The IR LED helped in transmission comparison.

The system successfully differentiated between pure water and contaminated samples using optical intensity differences. The results confirm that the proposed system can detect microplastics using simple optical sensing techniques. The performance of the system depends on proper alignment of LEDs and photodiode and stable sample placement..

5.3 IMPACT AND INNOVATION

The proposed system provides a low-cost alternative to expensive laboratory detection methods. It demonstrates how embedded systems and optical sensing can be used for environmental monitoring. The project shows innovation by combining multi-wavelength optical sensing with embedded processing in a compact design.

The system can be further improved by integrating machine learning algorithms for polymer classification and concentration estimation. Future improvements may also include IoT connectivity for remote monitoring and data logging. The project demonstrates a practical approach toward developing affordable environmental sensing devices.

5.4 SUMMARY

The experimental results confirm that the proposed optical sensing system can detect variations between pure water and contaminated water samples. The system successfully demonstrated real-time detection using simple hardware components. The next chapter presents the conclusion and future scope of the project.

CONCLUSION AND FUTURE SCOPE

Conclusion

The proposed Low-Cost Optical Sensor System for Microplastic Detection in Water was successfully designed and implemented. The system uses optical sensing principles such as transmission and scattering variation to detect the presence of microplastic particles. The experimental results showed that the system can differentiate between pure water and contaminated water based on optical intensity changes.

The use of simple components such as LEDs, BPW34 photodiode, LM358 amplifier, and ESP32 microcontroller makes the system economical and easy to implement. Compared to existing laboratory detection techniques, the proposed system provides a portable and real-time detection solution. The project demonstrates how embedded systems and optical sensing can be effectively combined for environmental monitoring applications.

Overall, the project proves that a simple optical sensing approach can be used as a preliminary detection method for identifying microplastic contamination in water.

Future Scope

The proposed system can be further improved in several ways to enhance performance and functionality.

- Machine learning algorithms can be added for polymer type classification
- Concentration estimation can be improved using calibration techniques
- Multi-angle photodiode arrangement can improve detection accuracy
- Compact PCB design can improve portability.
- Mobile application integration can be implemented
- Optical filters can be added for better wavelength isolation
- AI-based analysis can improve detection reliability

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